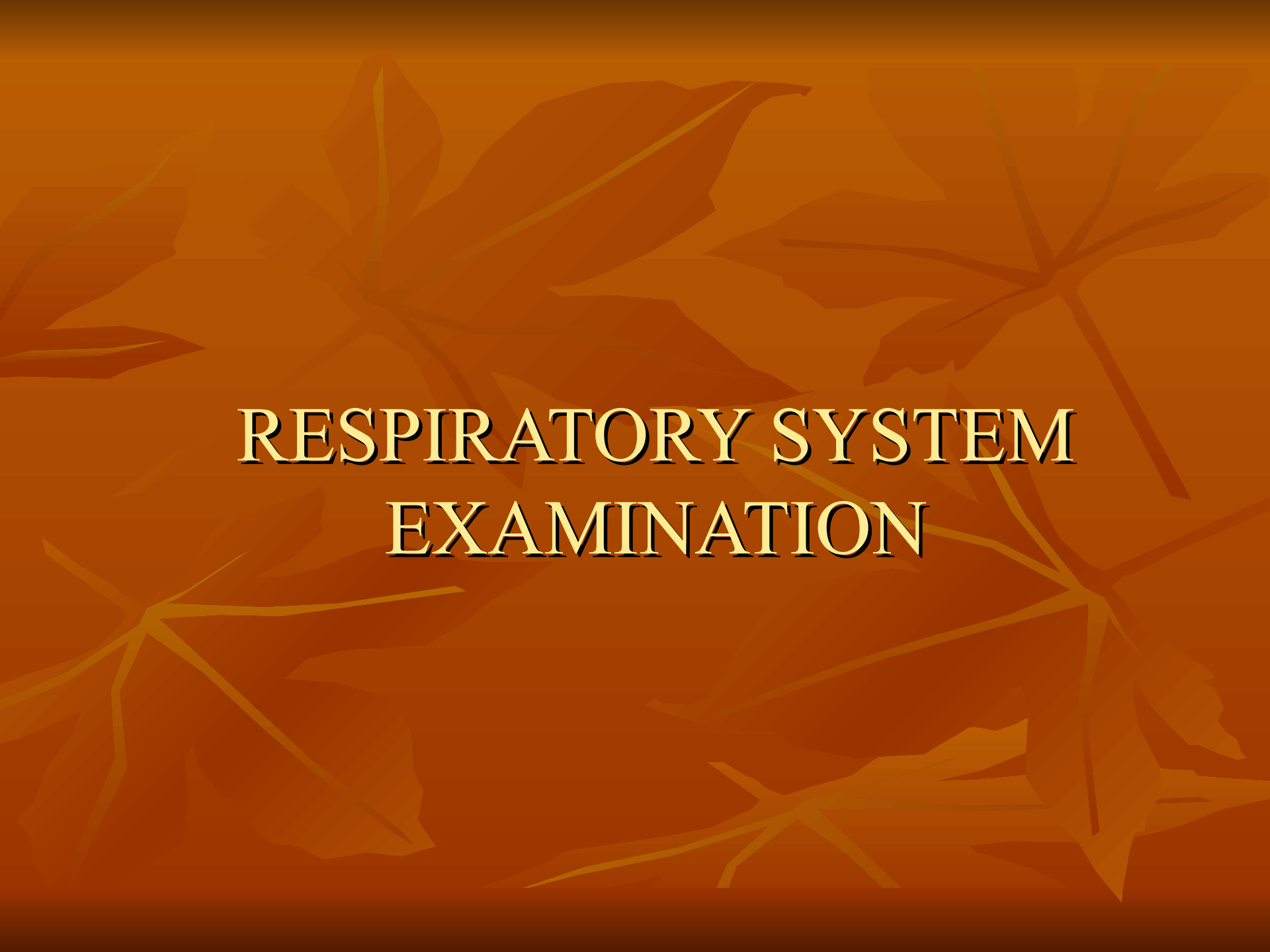
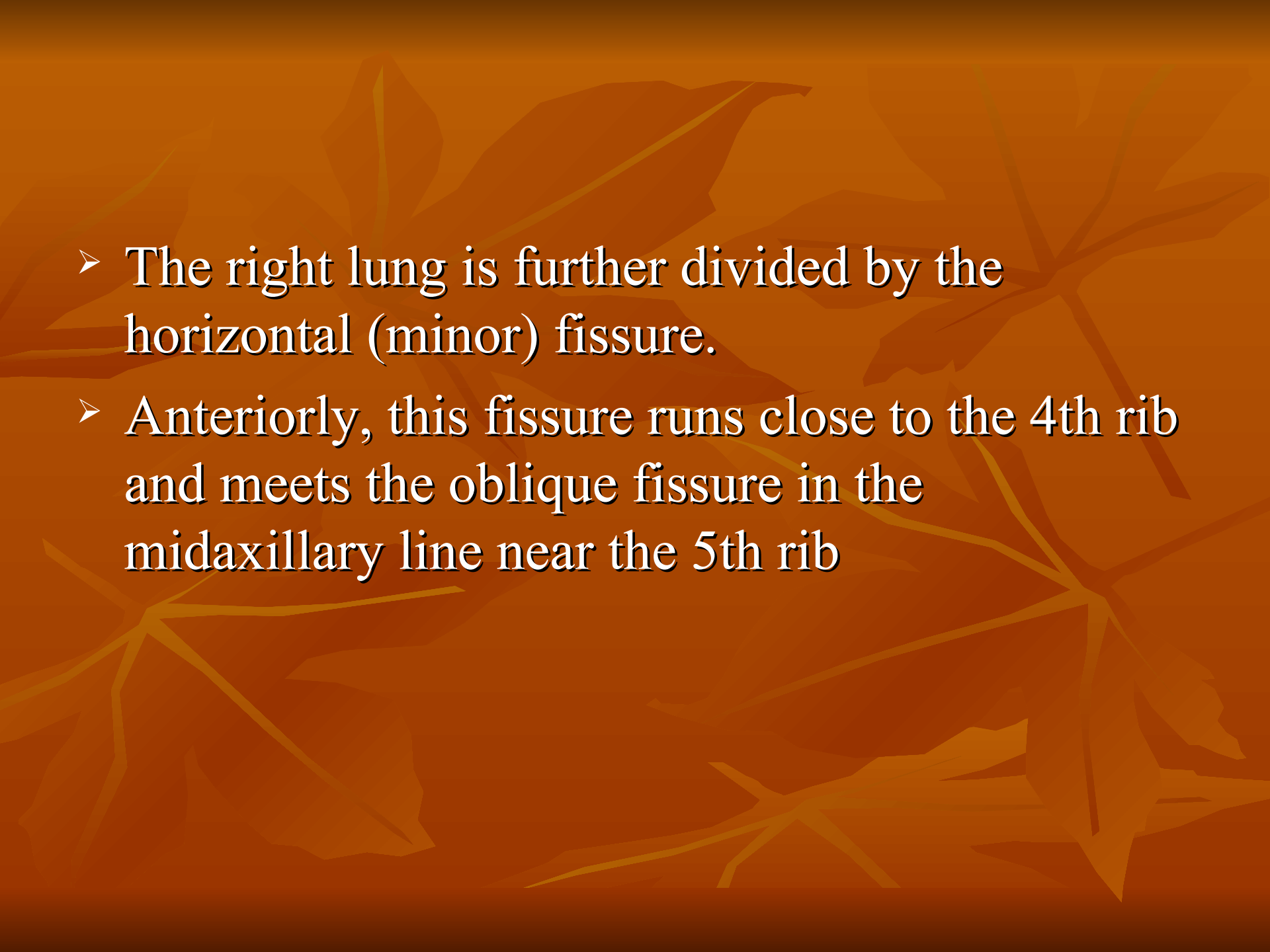


The background of the slide features a pattern of stylized, overlapping leaves in various shades of orange and brown, creating a textured, autumnal effect.

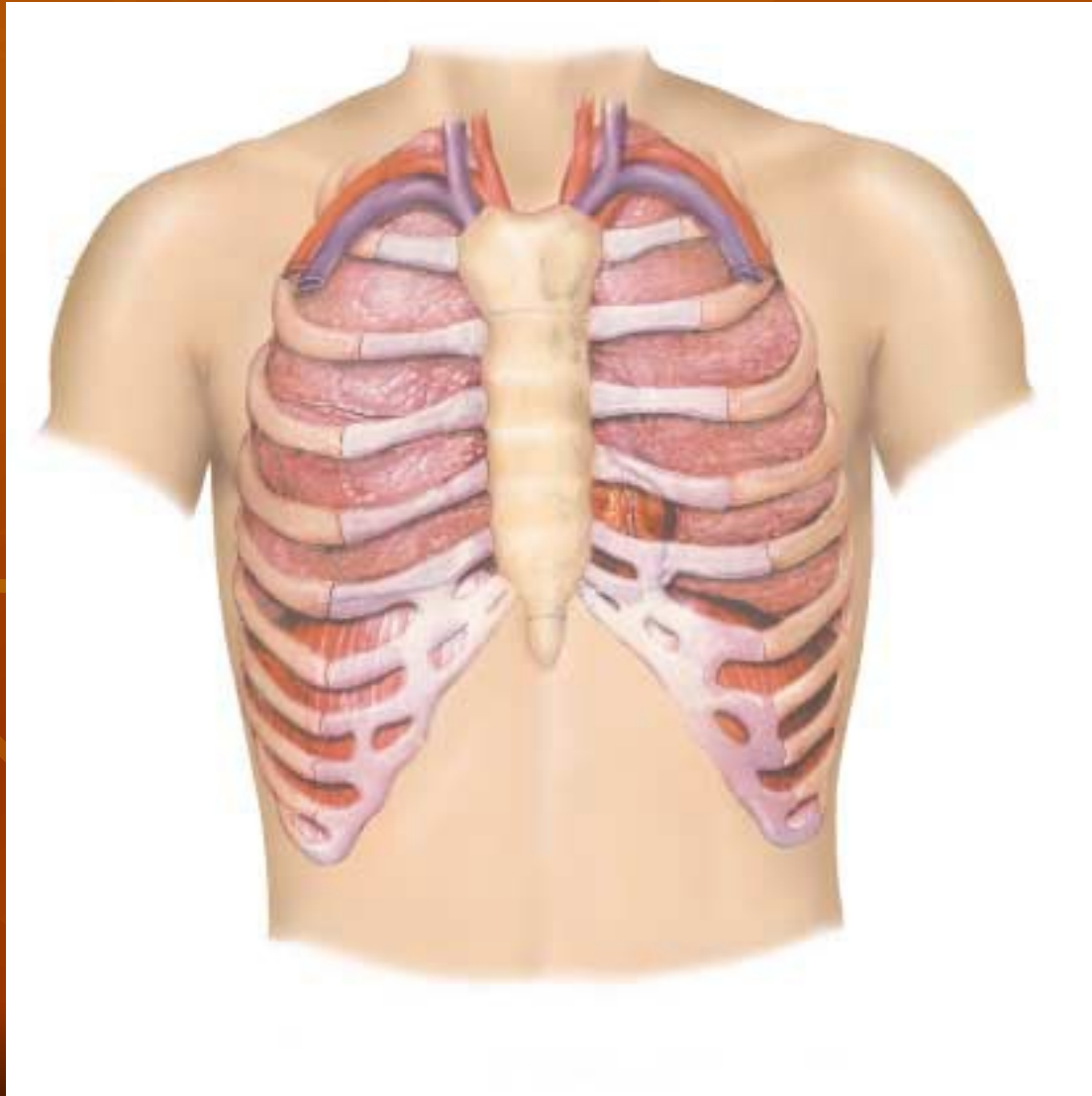
RESPIRATORY SYSTEM EXAMINATION

Lungs and lobes

- Anteriorly, the apex of each lung rises about 2- 4cm above the inner third of the clavicle.
- The lower border of the lung crosses the 6th rib at the midclavicular line and at the 8th rib at the midaxillary line and 10 rib at posterior scapular line.
- Each lung is divided about in half by an oblique (major) fissure. This fissure runs from the 3rd thoracic spinous process obliquely down and around the chest to the 6th rib at mid-clavicular line.

- 
- The right lung is further divided by the horizontal (minor) fissure.
 - Anteriorly, this fissure runs close to the 4th rib and meets the oblique fissure in the midaxillary line near the 5th rib

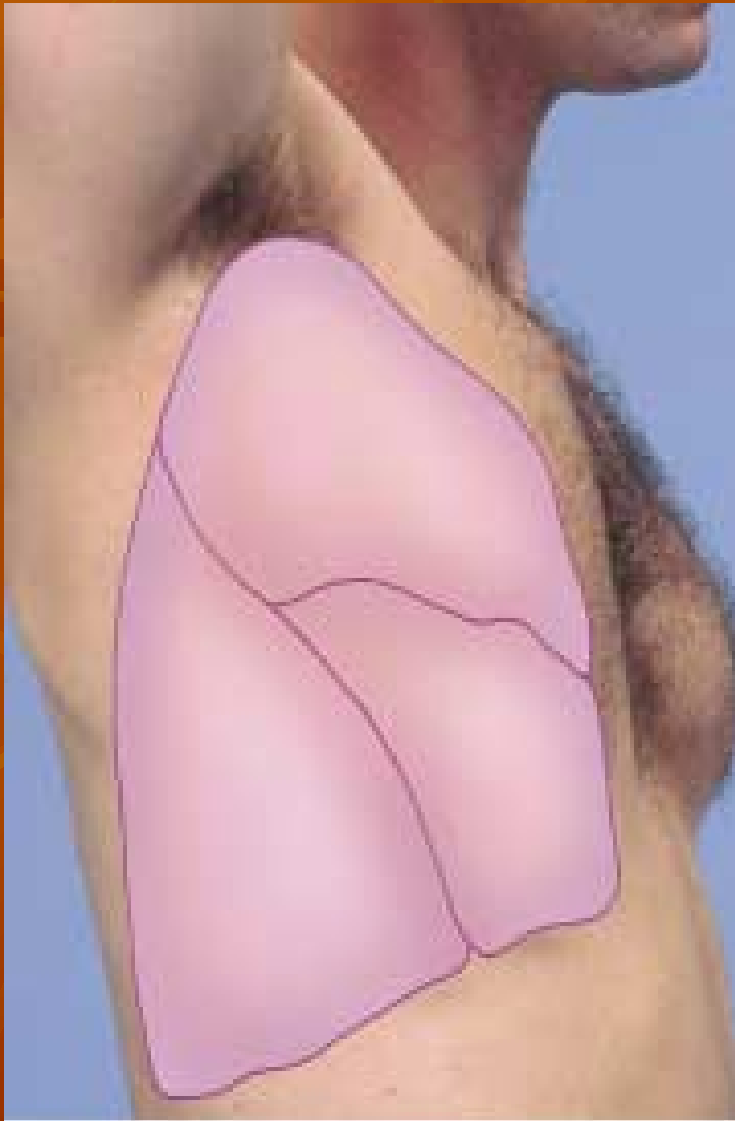
Land marks



Cont..



Cont..



Lung lobes



Anterior view



Right lateral view

Cont...



Posterior view



Left lateral view

Location of trachea & major bronchi

- The trachea bifurcates into its main stem bronchi at about the level of the sternal angle anteriorly and 4th thoracic spinous process posteriorly.

■ Breathing

- Breathing is largely an automatic act, controlled by the brainstem and mediated by muscles of respiration

Cont..



■ **Locate pulmonary findings** in external terms (location on the chest) such as:

- ❖ Supraclavicular – above the clavicle
- ❖ Infraclavicular –below the clavicle
- ❖ Interscapular –between the scapulae
- ❖ Infrascapular –below the scapula
- ❖ Bases of lungs –the lower most portions
- ❖ Axilla
- ❖ Upper, middle, & lower lung fields

The respiratory System examination

- ❖ Positioning the patient
- ❖ The patient should be undressed to the waist.
- ❖ If he or she is not acutely ill, the examination is easiest to perform with the patient sitting over the edge of the bed or even on a chair.

Respiratory System examination Con...

- The cardinal steps of chest examination are
Inspection,
Palpation,
Percussion and
Auscultation

Inspection

- ❖ **General Assessment**
 - ❖ **Physique**
 - ❖ **Cyanosis/Paler**
 - ❖ **Clubbing**
 - ❖ **Flaring of ala nasi**
 - ❖ **Breathing patterns**
 - ❖ **Use of accessory muscles**
 - ❖ **Respiratory rate and rhythm**
 - ❑ Normal=14-16/min
 - ❑ Tachypnoea > 20/min
 - ❖ **Chest indrawing (retractions)**
 - ❖ **Venous pulse**

Inspection of the Chest

- ❑ Appearance of the chest/Shape
 - Bilaterally symmetrical and an elliptical in cross section
 - Shape of the chest
 - Kyphosis
 - Scoliosis
 - Flattening
 - Over inflation
- ❑ Movement of the chest
 - ❑ symmetry
- ❑ Symmetry
- ❑ Observe the chest for –rate and rhythm
–chest expansion

Cyanosis

- Cyanosis is bluish discoloration of the skin &/ or mucus membrane caused by presence of excessive amount of reduced hemoglobin in capillary blood.
- Look central and peripheral cyanosis:- over tongue, lips, other mucous membranes & fingers.

Types of cyanosis

1. **Central cyanosis:-** results when O₂ saturation is below 80– 85% due to imperfect oxygenation of blood (deoxygenated blood mixes with oxygenated arterial blood in the heart, great vessels or lungs)

➤ **Causes:**

a). *Lung diseases such as*

- Severe hypoxic lung diseases such as
 - chronic obstructive airway disease (COAD),
 - severe pneumonia

- b). *Cardiac diseases*, such as
 - Heart failure
 - Lesions resulting in shunting from pulmonary to systemic circulation such as
 - Right to left shunting VSD known as Eisenmenger's syndrome
 - Patent ductus arteriosus (PDA)
 - Transposition of great arteries (TGA)
- **Affected areas:** characteristically affects tongue – the tongue is the best place to look for central cyanosis. Blue discoloration of lips, tongue, & other mucous membranes & nail beds.
- Cyanosis detected in the hands or nails is central if the hands are warm.

- **2. *Peripheral cyanosis*** (blue discoloration of arms, legs, face) - will occur in the above mentioned causes of central cyanosis,
 - but may also be induced by changes in the peripheral & cutaneous vascular system
 - in patients with normal O₂ saturation when local circulation is impaired & there is greater extraction of O₂ from the haemoglobin by the tissues – as in exposure to cold, in hypovolemia, and in arterial diseases.
 - Affected area:- Peripheral cyanosis is seen on hands & feet & these are usually cold

Clubbing of fingers

- Clubbing of fingers is the bulbous enlargement (like drum stick) of soft parts of the terminal phalanges due to
 - an increase in the vascularity of the distal fingers and consequently an increased sponginess of the nail beds
 - with over curving of the nails both transversely and longitudinally.
- Examine the nail beds for clubbing:

- Clubbing tends to affect the index finger first and it is this finger that should be examined.
- Observe for bulbous enlargement of the ends of the fingers. If it not obvious, feel for proximal flaccuation
- ('floating fingers') by exerting pressure over the proximal nail plate; and then observe the finger from the lateral aspect to assess the nail fold/nail plate angle which is normally an obtuse angle about 160

Grades of clubbing

- ❖ Grade I: Spongy, boggy feeling on pressing the nail bed – (early clubbing)
- ❖ Grade II: loss of angle at the nail bed >160 i.e. 180 or more .
- ❖ Grade III: widening of the distal part of the phalanx, spooning nail (late clubbing)=drum stick
- ❖ ***Causes of clubbing:-*** Clubbing is due to long standing lack of oxygen to the peripheral tissues



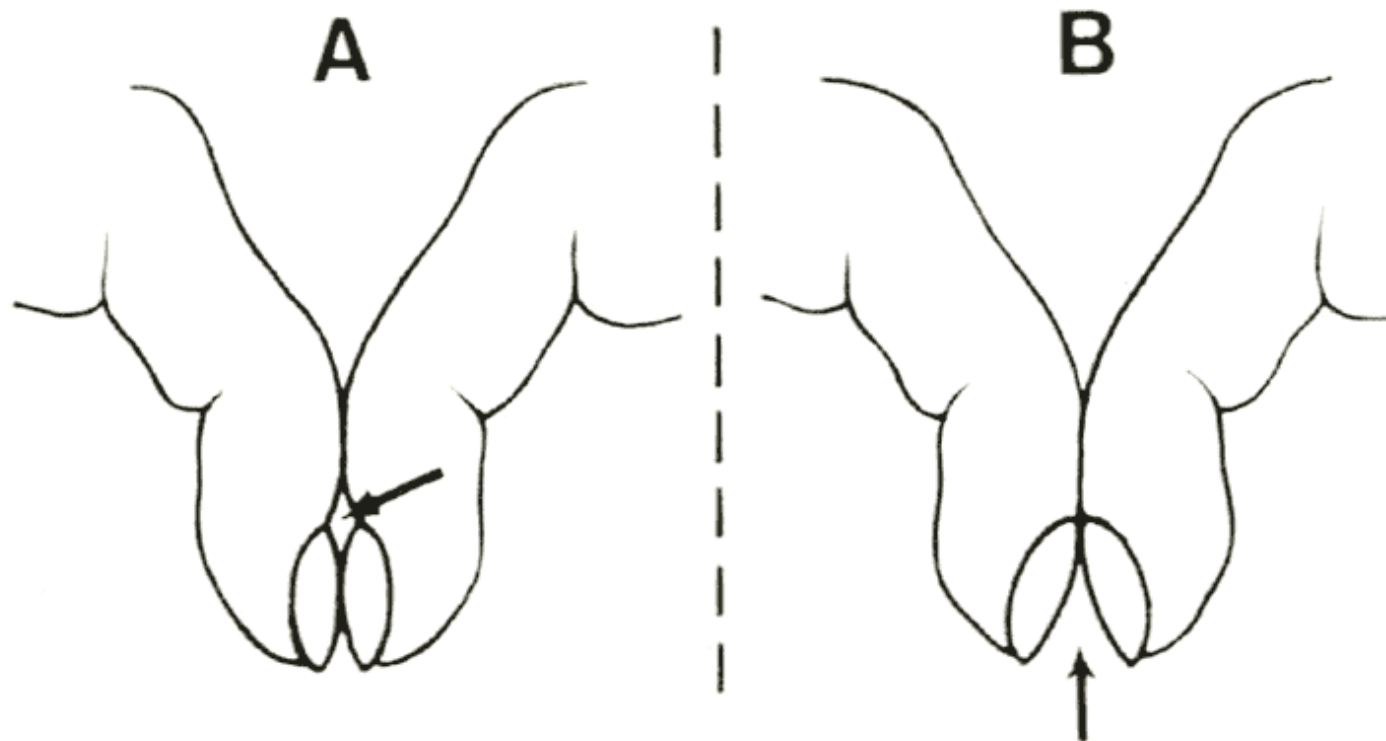


Fig. 1. Diagrams illustrating (A) normal finger contour, and (B) clubbing of the fingers.

Causes...

■ 1. Respiratory system

- Bronchial Ca
- Chronic lung suppuration, such as empyema, lung abscess, bronchiectasis
- Cystic fibrosis
- Fibrosing alveolitis
- Mesothelioma
- Carcinoma of lung, pulmonary Tb (lesser degree of clubbing)
- Chronic bronchitis is NOT a cause of clubbing

Cont....

■ 2. Cardiac disease

- Cyanotic congenital heart disease
- Infective endocarditis

■ 3. GI causes

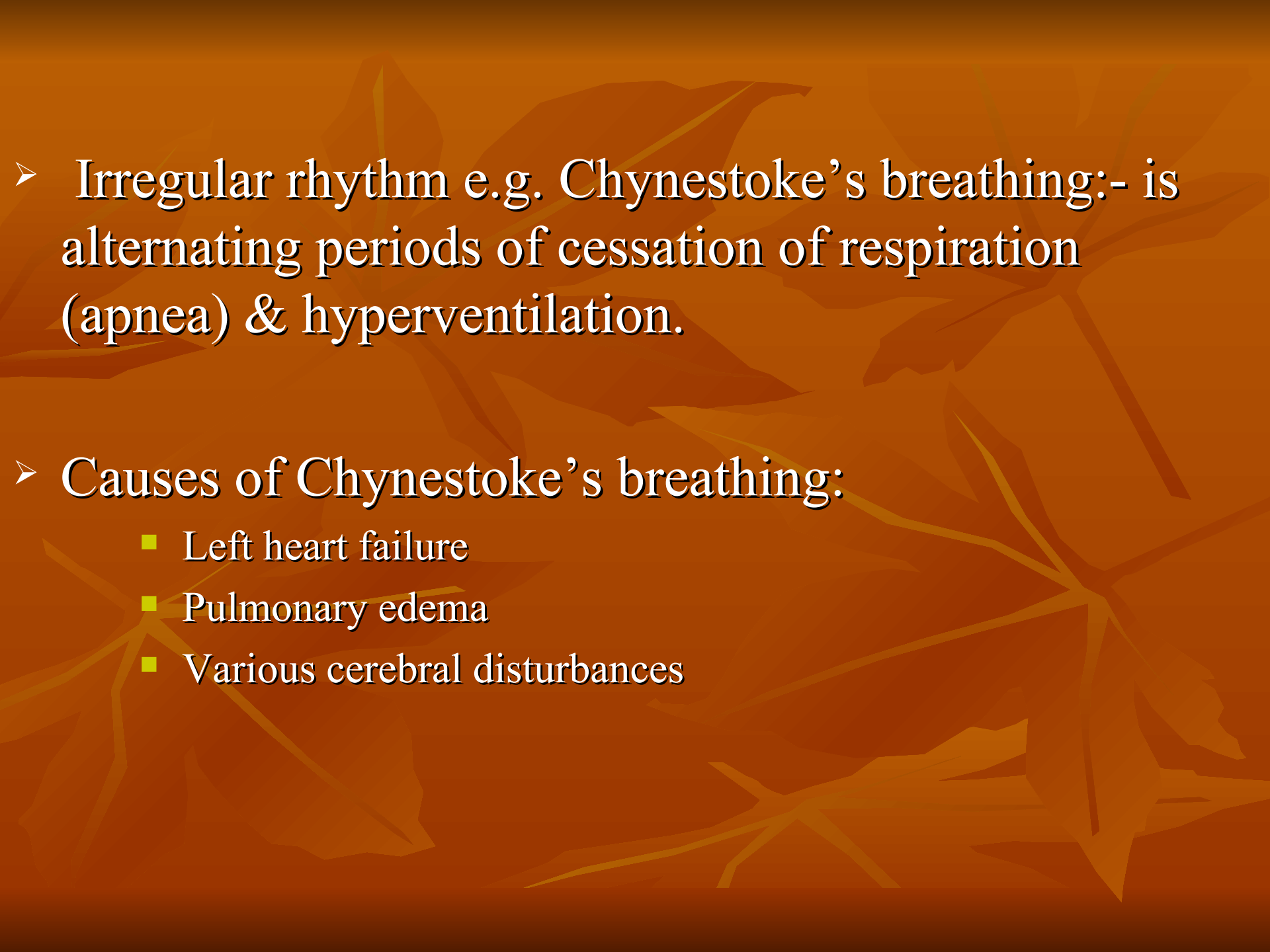
- Inflammatory bowel disease (esp. Crohn's disease, ulcerative colitis)
- Cirrhosis of the liver
- GI lymphoma
- Malabsorption (Coeliac disease)

- **Flaring of ala nasi**

- ✓ Flaring of ala nasi is the spreading out of the nostrils during expiration in children due to respiratory distress.

- **Breathing patterns (rate, rhythm, and depth)**

- The rate, rhythm, and depth of breathing should be noted carefully.
 - a. Rate:- see under vital sign, RR
 - b. Rhythm of breathing (regular, or irregular)

- 
- Irregular rhythm e.g. Chynestoke's breathing:- is alternating periods of cessation of respiration (apnea) & hyperventilation.
 - Causes of Chynestoke's breathing:
 - Left heart failure
 - Pulmonary edema
 - Various cerebral disturbances

- C). Depth of breathing (shallow, normal, deep)
 - i. Abnormal deep breathing
 - a). Deep sighing breathing

Kussmual breathing= rapid, deep breathing

Causes of rapid, deep breathing :

- Metabolic acidosis (e.g.diabetic keto-acidosis (DKA), uremia, pre-eclampsia, eclampsia) = acidotic breathing
- Pontomesencephalic lesions
- severe pneumonia
- Vigorous exercise & a state of anxiety

- **b). Forced expiration:-** a prolonged expiratory phase with visible use of accessory muscles of the neck & intercostals.
 - Occurs in asthma, chronic bronchitis, pulmonary emphysema

- **c). Forced inspiration:-** when the lung has become mechanically rigid as a result of fibrosis or pulmonary edema; or in blockage of the large airways such as trachea or larynx

- ii. Shallow, rapid breathing:- seen with anatomical defects, pulmonary infection, pleuritic disease, and metabolic disorders.
- iii. shallow, slow breathing may occur as a result of central nervous system pathology, metabolic disease, and drug effect.
- Signs of respiratory distress:
 - Flaring of ala nasi
 - Retractions at suprasternal notch, intercostal & subcostal regions
 - Use of accessory muscles of respiration
 - Cyanosis
 - Grunting

■ Use of accessory muscles

- Is there use of the neck muscles such as sternomastiod, scaleni & trapezius muscles for respiration? (Yes/No)
- The normal muscles of respiration are the diaphragm , the intercostal muscles and abdominal muscles
- Diaphragm descends on inspiration & cause outward abdominal movement.
- Intercostal muscles cause chest expansion on inspiration.
- In labored breathing (i.e. in respiratory distress) neck muscles (accessory muscles of respiration) are recruited to lift the chest

■ Shape of the chest wall

- Inspect the chest wall for deformities (first the front of the chest and then the back).
- Normal chest wall – is symmetrical and,
- In the infant or young child, almost round
- The transverse diameter increases with age, thus elliptical in cross section
- Abnormalities (deformities) of chest wall includes:
- Barrel chest – a persistently round (increased antero-posterior) diameter of chest wall.
 - Cause:-chronic hyperinflation (e.g. in severe asthma, chronic obstructive airway disease (COAD))
 - such as cystic fibrosis or chronic asthma, emphysema

- Pigeon chest - is chest wall with prominent sternum & flat chest (pectus carinatum), is sequel of chronic respiratory disease in childhood.
- Funnel chest - is chest wall with local sternum depression at lower end (pectus excavatum).
- Kyphosis – is forward bending of spines
- Scoliosis –is lateral curvature of spines
- Harrison's groove—is a horizontal depression along the lower border of the chest that corresponds with the costal insertions of the diaphragm—is sign of rickets.
- Rachitic rosaries—is a palpable or visible costo-chondrial beading – is sign of rickets

■ **Symmetry of chest wall movement**

- Inspect movements of the two sides & both upper & lower parts of the chest.
- Inspiration normally results in expansion of the chest wall and depression of the diaphragm. Normal chest moves symmetrically & equal on both sides.
- Impairment of respiratory movement on one or both sides or unilateral lag (or delay) in that movement
- suggests disease of the underlying lung or pleura on affected side – such as pneumonia, pleural effusion, pneumothorax, lung collapse, atelectasis, or unilateral bronchial obstruction or a foreign body lodged in one of the mainstem bronchi

- Asymmetrical (unilateral) chest movement—diminished chest movement occurs in lung or pleural lesions. e.g. on the side of pneumothorax, extensive consolidation (e.g. lobar pneumonia).
- Bilateral restricted chest movement is noted in chronic obstructive emphysema.
- Paradoxical breathing can be seen in patients with neuromuscular disease – if during inspiration the diaphragm rise and the chest wall and abdomen collapse on the involved side

- **Chest indrawing (retractions)**
 - Retraction (indrawing) of the intercostal spaces, subcostal, suprasternal, and supraclavicular fossae during
 - inspiration is present in airway obstruction & non-compliant lung

■ Chest expansion measurement

- Chest expansion can be measured with tape meter around the chest at about the level of the nipples or 4th intercostals space in males, or just below the breasts in females on deep maximum inspiration and on maximal forced expiration. Take the difference between these two measurements.
- In children, normally it is 2cm
- In a fit young man, the chest may expand $> 5\text{cm}$ (ranges 5–8 cm)
- In severe emphysema, it may expand less than 1cm

Palpation

- ❖ **Tenderness**
- ❖ **Mass or swelling**
- ❖ **Position of trachea**
- ❖ **Tactile fremitus**
- ❖ **Chest expansion**

■ Tenderness

- Palpate the chest wall where patient complains of pain.
- Intercostal tenderness may be due to inflamed pleura (e.g tuberculosis).
- Causes of chest pain & tenderness:
 - Recent injury of the chest or inflammatory conditions
 - Intercostal muscular pain
 - Rib fracture
 - malignant deposits in the ribs
 - Herpes zoster before appearance of eruption
 - Pleurisy (inflammation of pleura)

■ **Mass /swelling**

- Determine nature of any mass or swelling with:
 - Site
 - Temperature
 - Tenderness
 - Size
 - Consistency
 - Surface
 - Mobility, etc.

■ **Position of trachea**

- Position of trachea indicates the position of upper mediastinum. Displacement of the trachea with displacement of apex beat with normal heart size may suggest mediastinal displacement.
- Normally on midline, may slightly deviates to the right.
- Abnormal tracheal deviations
 - Deviation to same side of the cause (pulled to one side), as in
 - Lung collapse
 - Lung fibrosis
 - Deviation to the opposite side of the cause (pushed to opposite side) by
 - Pleural effusion
 - Pneumothorax
- Note: - in lung consolidation no tracheal deviation occurs

■ **Tactile fremitus (TF)**

- TF refers to palpable vibrations transmitted through the broncho-pulmonary tree from the larynx to the surface of the chest wall when the patient speaks.
- 1. Ask the patient to say the following several times in a normal voice:
 - Ninety nine for English speakers
 - ‘arba arat’ for Amharic speakers
- 2. Palpate & compare symmetrical areas of both sides of the posterior, anterior and the lateral chest areas including the apices –for presence or absence & symmetry of TF

- Locate the area where TF increased, decreased or absent.
- Increased TF in
 - Lung consolidation
 - Lung fibrosis
- Decreased to absent TF when transmission of vibrations from the larynx to the surface of the chest is impeded by:
 - Obstructed bronchus
 - Chronic obstructive pulmonary disease (COPD)
 - Separation of the lung from chest wall by:
 - Pleural air e.g. Pneumothorax
 - Pleural fluid e.g. pleural effusion, hemothorax
 - Pleura thickening

■ Chest expansion

- Place the fingertips of both hands on either side of the lower rib cage so that the tips of the thumbs meet in the mid line (done either on anterior or posterior side of chest), then the patient is asked to breath deeply.
- Posteriorly, at the level of and parallel to the 10th ribs.
- If one thumb remains closer to the mid line – indicates that there is diminished expansion of the chest on that side.
- Causes:– see under symmetry of chest movement, above

Cont..



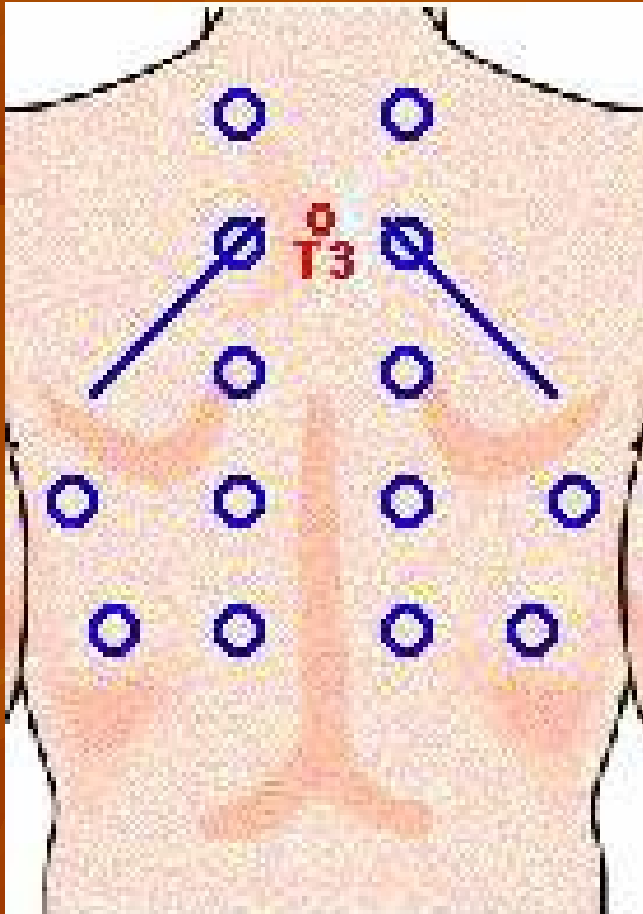
Percussion

- ❖ **Resonance**
- ❖ **Hyper resonance**
- ❖ **Dull**
- ❖ **Stony (flat) dullness**
- ❖ **Diaphragmatic excursion**

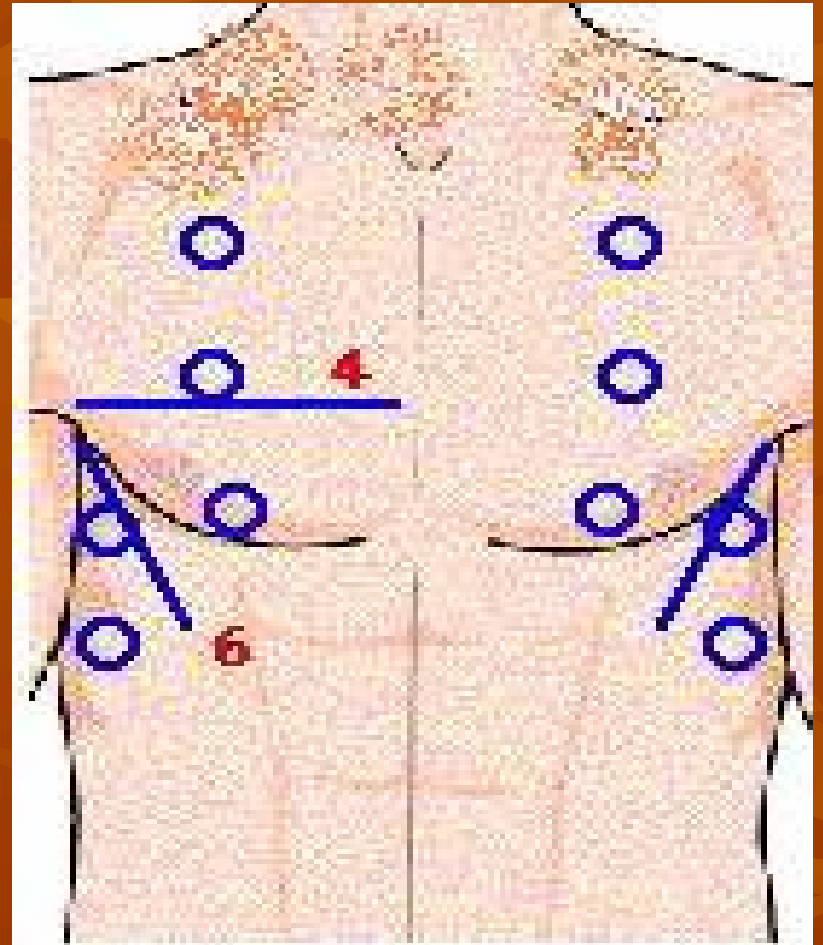
Percussion



Cont..



Posterior



Anterior

Proper Technique

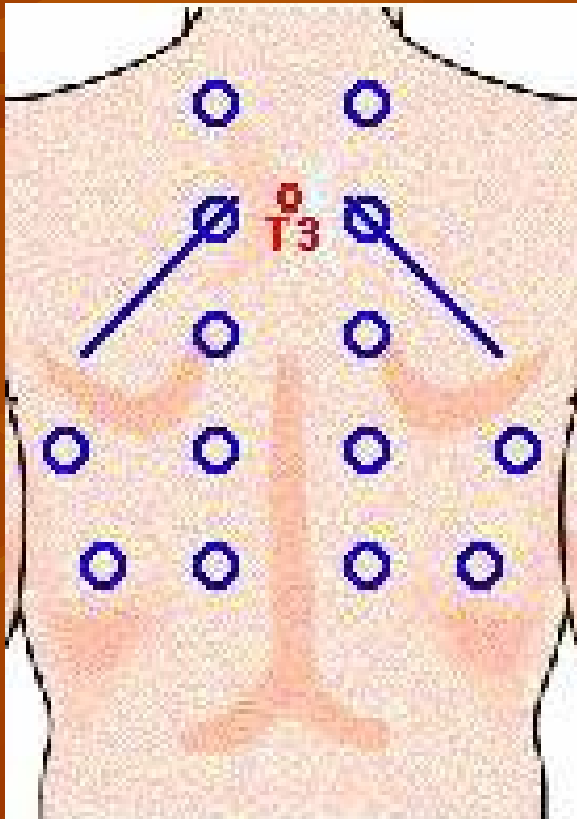
- Hyperextend the middle finger of one hand and place the distal interphalangeal joint firmly against the patient's chest.
- With the end (not the pad) of the opposite middle finger, use a quick flick of the wrist to strike first finger.
- Categorize what you hear as normal, dull, or hyperresonant.
- Practice your technique until you can consistently produce a "normal" percussion note on your (presumably normal) partner before you work with patients.

- Percuss symmetrical (equivalent) areas of both sides (including apices, posterior, lateral, & anterior) of the chest at about 5cm intervals from the upper to the lower chest (moving from left to right & right to left) & compare both areas –for relative resonance or dullness of the tissue underlying the chest wall.
- Posterior Chest
- 1. Tell the patient to cross his/her hands in front of their chest grasping the opposite shoulders so as to pull the scapulae laterally.

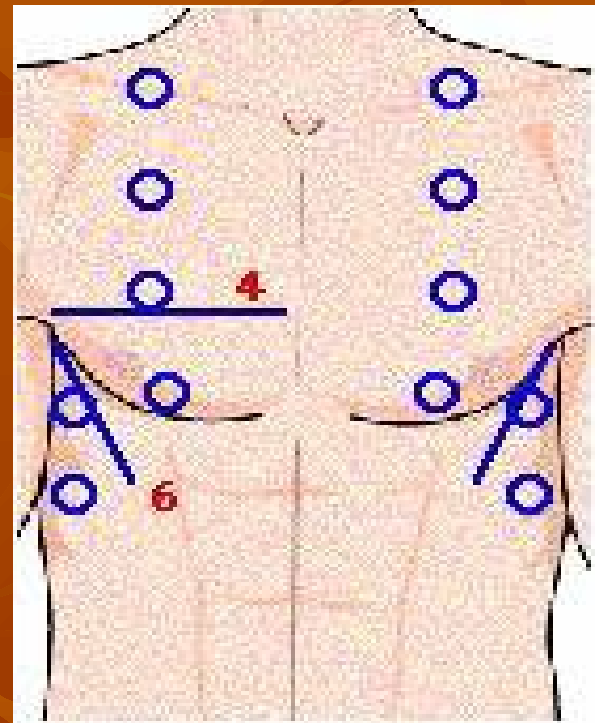
- 2. Percuss from side to side and top to bottom using the pattern shown in the illustration. Omit the areas covered by the scapulae.
- 3. Compare one side to the other looking for asymmetry.
- 4. Note the location and quality of the percussion sounds you hear.
- 5. Find the level of the diaphragmatic dullness on both sides

Cont..

(a) Posterior



(b) Anterior



Anterior Chest

- 1. Percuss from side to side and top to bottom using the pattern shown in the illustration.
- 2. Compare one side to the other looking for asymmetry.
- 3. Note the location and quality of the percussion sounds you hear
- The normal percussion note of underlying air-containing normal lung field is resonance

- The normal percussion note of underlying air-containing normal lung field is resonance.
- Abnormal percussion notes are:
 - Hyper resonance – occurs in emphysema (hyper inflated lung), or Pneumothorax (when pleural cavity is filled with air)
 - Dull as in lung consolidation, lung fibrosis, lung collapse, pleural effusion, hemothorax, pleural thickening, etc.
 - Stony (flat) dullness – is due to fluid or blood in the pleural cavity as in pleural effusion & hemothorax

Percussion Notes and Their Meaning

- ❖ Flat or Dull ----Pleural Effusion or Lobar Pneumonia
- ❖ Resonant---Normal Healthy Lung or Bronchitis
- ❖ Hyperresonant-- Emphysema or Pneumothorax

- **Diaphragmatic excursion**
- 1. Percuss along the scapular line on one side until the level of the diaphragmatic dullness.
- 2. Ask the patient to inspire deeply and hold his breath in.
- 3. Proceed to percuss down from the marked point – to determine the diaphragmatic excursion in deep
- inspiration.
- 4. Repeat the procedure on the opposite side.
- 5. Measure the distance between the upper & lower points in cm on each side.
- Excursion is normally 3–5cm bilaterally (symmetrically)

Cont..



Auscultation

- ❖ **Air entry**
- ❖ **Breath sounds**
- ❖ **Added (adventitious) sounds**
- ❖ **Coin sound**

- An organized symmetrical approach to auscultation of the lung fields should be used.
- Ask the patient to breathe (i.e. to let the air into the lungs & let it out again) while auscultating the chest.
- Sequential examination proceeds from one side of the chest to the other, comparing breath sounds in anatomically similar areas of both sides of posterior, lateral & anterior chest using the pattern shown in the illustration above. Omit the areas covered by the scapulae.

- Normal breath sounds are
 - over the lung tissue is called vesicular breath sound;
 - over the trachea is bronchial breath sound; and
 - between the two over main bronch is vesiculo-broncheal breath sound. .
- Ordinarily, deep mouth breathing produces clear, soft breath sounds over the lungs
- Auscultate the chest for both the quality and intensity of the breath sounds and for the presence of extra, or adventitious, sounds



■ **Air entry**

- ❖ Normal
- ❖ Decreased / absent
- ❖ Increased

➤ Normal air entry - in normal lung

➤ Decreased to absent air entry in

- Pleural effusion
- Lung collapse
- Pneumothorax
- Hemothorax
- Severe asthma
- Major bronchial obstruction

➤ Increased air entry in

- Lung consolidation
- Lung fibrosis

Breath sounds

- ❖ Vesicular breath sound
- ❖ Bronchial breath sound
- ❖ Amphoric breath sounds
- ❖ Decreased/absent
- ❖ Vocal resonance

Vesicular breath sound

- It is the breath sound heard over the normal lung parenchyma.
- It is rather quite low-pitched rustling sound without distinct pause (gap) between the end of the inspiration and the beginning of expiration.
- Vesicular breath sound inspiration phase greater than expiration

Broncho-vesicular sounds

- Normally heard in areas of the major bronchi especially at the apex of the right lung & the sternal border.
- ***Bronchial breath sound (BBS)***
- It is normally heard over the trachea.
- Shift of vesicular to bronchial breath sound over the lung tissue indicates pathology, lung consolidation.
- It is a harsh, tubular, sound, becomes inaudible just before the end of inspiration, so that there is a gap before the expiratory sound is heard.
- The expiratory sound lasts for most of the expiratory phase

- Bronchial breath sound (BBS) is heard over the lung fields in
 - Lung consolidation
 - Lung fibrosis
 - Over top of pleural effusion
- Amphoric breath sounds
 - It is a sound heard like that made by blowing over the mouth of a narrow necked glass e.g. bottle.
 - It is heard over:
 - Cavitory lesions
 - Top of pleural effusion

➤ *Decreased or absent breath sounds*

- Decreased or absent breath sounds can occur in:
 - Any condition that causes the deposition of foreign matter (air, fluid, blood) in the pleural space such as
 - pneumothorax, pleural effusion, hemothorax
 - Emphysema,
 - Endobronchial obstruction

Vocal resonance

- Tell the patient to speak normally ('one-one-one', ninety nine, etc.) while auscultating the chest wall.
- Normal speech is muffled and indistinct when heard at the chest wall through normal lung tissue.
- Normal speech is heard clearly through consolidated lung (vocal resonance)

Bronchophony

- 1. Ask the patient to say "ninety-nine", or "arba arat", several times in a normal voice.
- 2. Auscultate several symmetrical areas over each lung.
- 3. The sounds you hear should be muffled and indistinct. But if Louder, clearer sounds are called bronchophony.

Whispered Pectoriloquy

- 1. Ask the patient to whisper "ninety-nine", or "arba arat", several times.
- 2. Auscultate several symmetrical areas over each lung.
- 3. You should hear only faint sounds or nothing at all. If you hear the sounds clearly this is referred to as
 - whispered pectoriloquy.

Egophony

- 1. Ask the patient to say "ee" continuously.
- 2. Auscultate several symmetrical areas over each lung.
- 3. You should hear a muffled "ee" sound. If you hear an "ay" sound this is referred to as "E → A" or
 - Egophony.

Added (adventitious) sounds

- ❖ Crackles / rales
- ❖ Wheezes/ronchi
- ❖ Pleural friction rub
- ❖ Stridor

- Atypical (added, adventitious) sounds are not alterations in breath sounds but superimposed on breath sounds.
- Added breath sounds in the form of rales, rhonchi, or wheezes are heard most often in patients with underlying pulmonary disease.
- Upper airway congestion often produces coarse sounds that, when transmitted through the larger airways, may give the impression of an underlying lung abnormality.
- Frequently, coughing and/or vigorous crying will clear a congested upper airway, helping the clinician distinguish between upper and lower airway sounds.
- Includes

- Rales / crepitations / crackles: (rales are old terms)
- Rales/crepitations are short, discrete, interrupted crackling sound that are heard during inspiration.
- Fine crepitation is heard in
 - pulmonary edema
 - fibrosing alveolitis
- Coarse crepitation is heard in
 - bronchiectasis
 - bronchogenic pneumonia

- ***Rhonchi***
- Rhonchi are continuous sounds produced by the movement of air in the presence of free fluid in the airway lumen, the tracheobroncheal tree. High pitched, sibilant rhonchi are called wheezes.
- **Wheezes:** are often audible at the mouth as well as through the chest wall.
- Wheezes, which are generally more prominent during expiration than inspiration, reflect the oscillation of airway walls that occurs when there is airflow limitation, as may be produced by bronchospasm, airway edema or collapse, or intraluminal obstruction by neoplasm or secretions



➤ Wheeze is heard in:

➤ Bronchial asthma

➤ Bronchitis

➤ Laryngeal spasm

➤ Tracheal fibrosis

➤ Congestive heart failure (cardiac asthma)

Pleural friction rub

- Pleural friction rub is heard as creaking noise liked to that emitted by compression of new leather .
- It indicates inflamed pleural surfaces rubbing against each other, often during both inspiratory and expiratory phases of the respiratory cycle.
 - e.g. inflammatory conditions of the pleura (pleurisy) from adjacent pneumonia or Tb, pulmonary infarction

Stridor

- Stridor is a wheeze that is entirely or predominantly inspiratory i.e. heard on inspiration and arises from a narrowed airway outside the thorax (usually trachea) that tends to close on inspiration
- In small children, an inspiratory high-pitched stridorous sound with or without significant respiratory distress may be the result of narrowing at or near the larynx or anywhere along the trachea.

This condition can be caused by a croup-like illness, anatomical defect, mass lesion, foreign body, or external obstruction.

Patients in severe respiratory distress with stridor may have epiglottitis

Comparison of the chest signs in common respiratory

Disorder	Mediastinal displacement	Chest wall movement	Percussion note	Breath sounds	Added sounds
Consolidation	None	Reduced over affected area	Dull	Bronchial	Crackles
Collapse	Ipsilateral shift	Decreased over affected area	Dull	Absent or reduced	Absent
Pleural effusion	Heart displaced to opposite side (trachea displaced only)	Reduced over affected area	Stony dull	Absent over fluid; may be bronchial at upper border	Absent; pleural rub may be found above effusion

Pneumothorax	Tracheal deviation to opposite side if under tension	Decreased over affected area	Resonant	Absent or greatly reduced	Absent
Bronchial asthma	None	Decreased symmetrically	Normal or decreased	Normal or reduced	Wheeze
Interstitial pulmonary fibrosis	None	Decreased symmetrically (minimal)	unaffected by cough or posture	Normal	Fine, late or pan-inspiratory crackles over affected lobes

The background of the slide is a solid orange color with a pattern of stylized, darker orange leaves. The leaves are arranged in a way that they appear to be floating or falling, with some overlapping. The overall effect is a warm, autumnal feel.

Thank you!